CLOUDOPS GUARDIAN

Intelligent Cloud Infrastructure Monitoring & Auto-Healing Platform

Created and Developed by Hitesh Patil

Project documentation | Revised October 2026

Project domain	Cloud / DevOps / Infrastructure Monitoring
Cloud platform	Amazon Web Services (AWS)
Core services	EC2, VPC, CloudWatch, Lambda, Systems Manager, EventBridge Scheduler, SNS
Web service	NGINX on Amazon Linux 2023
Project objective	Detect service health issues, attempt automated recovery, monitor infrastructure, and notify on detected failures.
Creator	Hitesh Patil

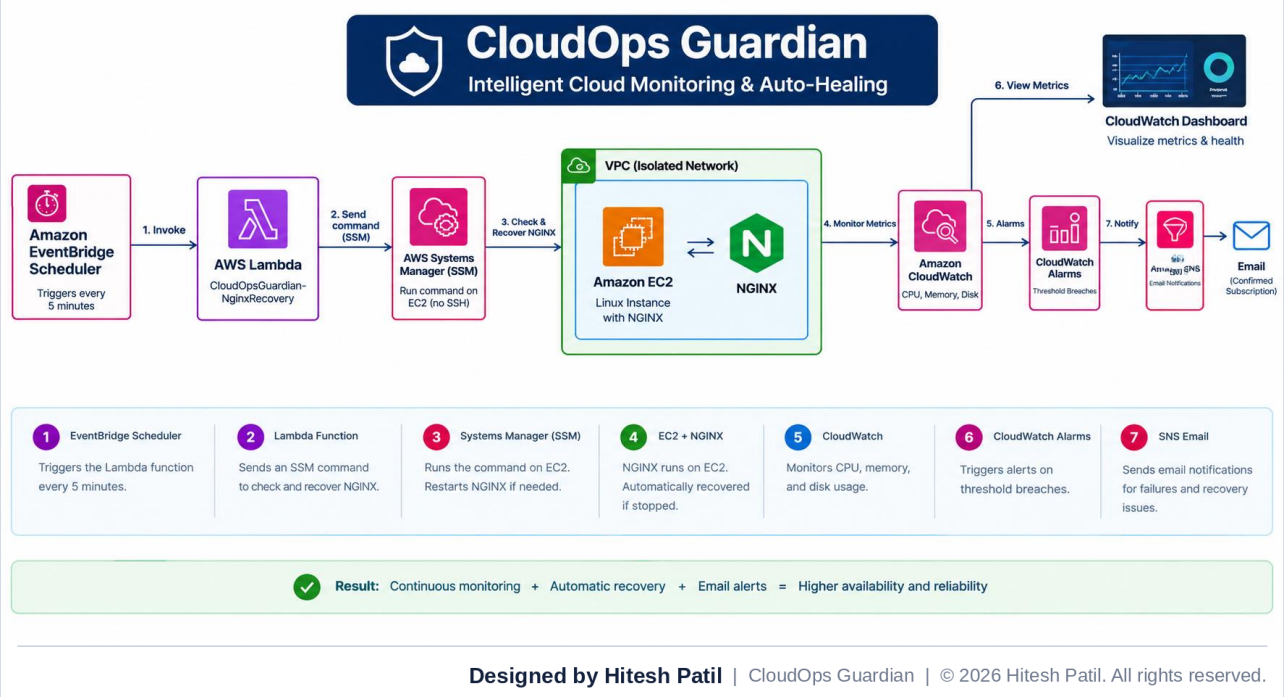


Figure 1. CloudOps Guardian architecture overview. Designed by Hitesh Patil.

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1. Project Overview

CloudOps Guardian is a hands-on AWS project that monitors an EC2 instance and its NGINX web service. CloudWatch collects infrastructure metrics, alarms monitor thresholds, EventBridge Scheduler invokes a Lambda function on a five-minute schedule, and Lambda uses AWS Systems Manager Run Command to check NGINX and attempt a restart if it is not active.

The system also uses Amazon SNS to send email alerts for selected command/recovery failure paths. The SNS email subscription has been confirmed, and a manual SNS publish test successfully delivered an email. End-to-end email delivery from a deliberately induced Lambda recovery failure still needs separate verification.

Project goals

- Monitor CPU, memory, and disk usage.
- Detect whether NGINX is active.
- Restart NGINX when the health-check command detects it is stopped.
- Run checks on a schedule without relying on SSH for remote command execution.
- Display metrics on a CloudWatch dashboard.
- Send email notifications for detected failures.

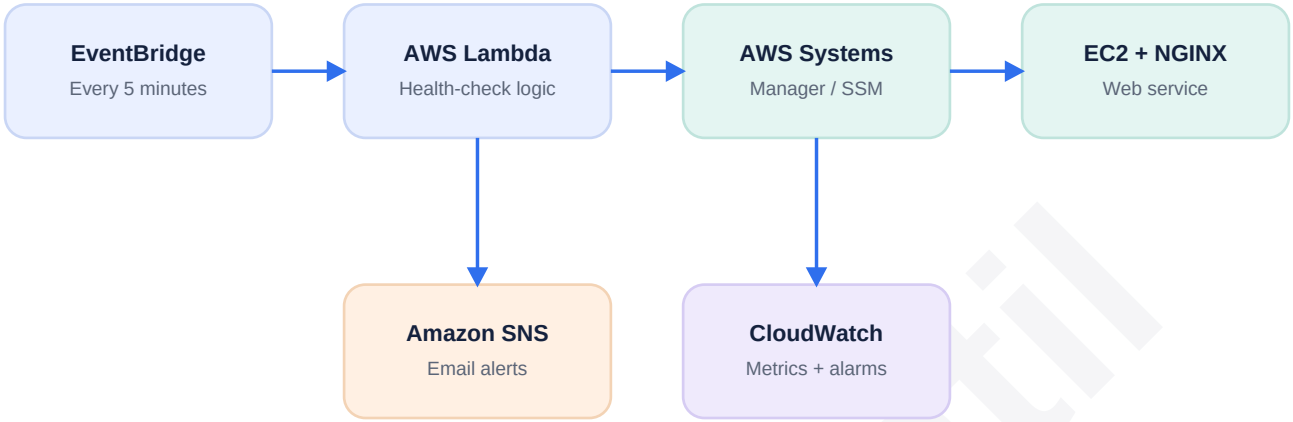
Current validation status

Area	Known result	Status
Lambda health check	Returned `NGINX_HEALTHY` with status Success	**Tested**
NGINX recovery	Earlier output showed `NGINX_DOWN_RESTARTING` and `NGINX_RECOVERY_SUCCESS`	**Tested**
CloudWatch metrics	CPU, memory, and disk metrics configured; dashboard widgets added	**Configured**
SNS email delivery	Manual SNS publish test delivered an email	**Tested**
Failure email via Lambda	Failure-handling code is present; end-to-end failure email not yet demonstrated	**Needs verification**

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2. Architecture

The scheduled workflow and monitoring components are shown below. CloudWatch metrics and alarms operate alongside the recovery workflow; SNS is used for notifications.



Designed by Hitesh Patil

Flow: scheduled check → Lambda → SSM command → NGINX status/recovery

Main workflow

1. EventBridge Scheduler invokes the Lambda function every five minutes.
2. Lambda calls Systems Manager `SendCommand` using `AWS-RunShellScript` for the managed EC2 instance.
3. The command checks `systemctl is-active nginx`.
4. If NGINX is active, the command reports `NGINX_HEALTHY`.
5. If NGINX is inactive, the command attempts a restart and verifies the service status.
6. Lambda polls the SSM invocation result and publishes an SNS message for configured failure conditions.

AWS resources used

Resource	Purpose	Configured details
VPC	Network boundary	cloudops-vpc; CIDR 10.0.0.0/16
Public subnet	Hosts EC2 in public network	cloudops-public-1a; 10.0.1.0/24
EC2	Runs Amazon Linux 2023 and NGINX	t3.micro; managed by SSM
CloudWatch / CWAgent	Metrics, alarms, dashboard	CPUUtilization, mem_used_percent, disk_used_percent
Lambda	Health check and recovery orchestration	CloudOpsGuardian-NginxRecovery
EventBridge Scheduler	Periodic trigger	CloudOpsGuardian-Every5Minutes
SNS	Email notifications	CloudOps-Guardian-Alerts

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3. Monitoring and Alarms

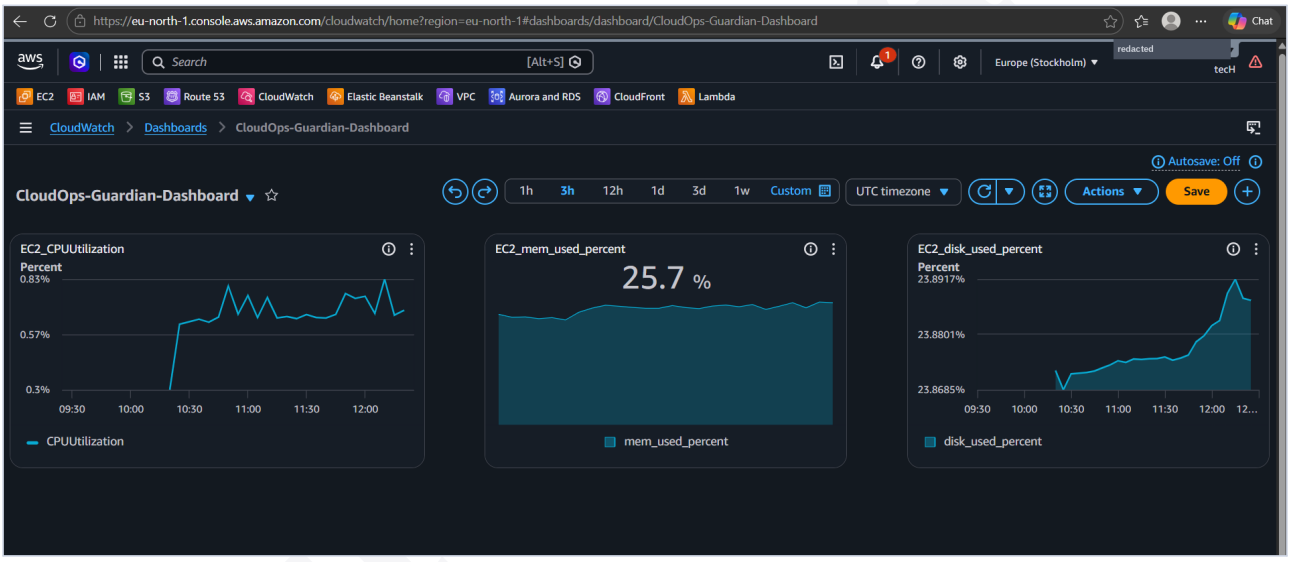
CloudWatch Agent publishes memory and disk metrics under the CWAgent namespace. EC2 CPUUtilization is available as an EC2 metric. CloudWatch alarms were created for high CPU, high memory, and high disk usage.

Configured alarms

Alarm	Metric	Threshold	Purpose
CloudOps-Guardian-High-CPU	CPUUtilization	≥ 80%	Detect high CPU utilization
CloudOps-Guardian-High-Memory	mem_used_percent	≥ 80%	Detect high memory use
CloudOps-Guardian-High-Disk	disk_used_percent	≥ 80%	Detect high root disk use

Dashboard

The CloudWatch dashboard is named `CloudOps-Guardian-Dashboard` and contains line widgets for CPU utilization, memory utilization, and disk utilization. These graphs help compare metric trends over a selected time range.



CloudWatch dashboard: CloudOps-Guardian-Dashboard (account details redacted).

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4. Auto-Healing and Lambda Logic

The Lambda function uses boto3 clients for Systems Manager and SNS. It sends an SSM command to the configured instance, waits for the command result, and inspects the output and status.

Key output values

- `NGINX_HEALTHY` — NGINX was active when checked.
- `NGINX_DOWN_RESTARTING` — NGINX was inactive and restart was attempted.
- `NGINX_RECOVERY_SUCCESS` — restart and status verification succeeded.
- `NGINX_RECOVERY_FAILED` — the restart or verification failed.

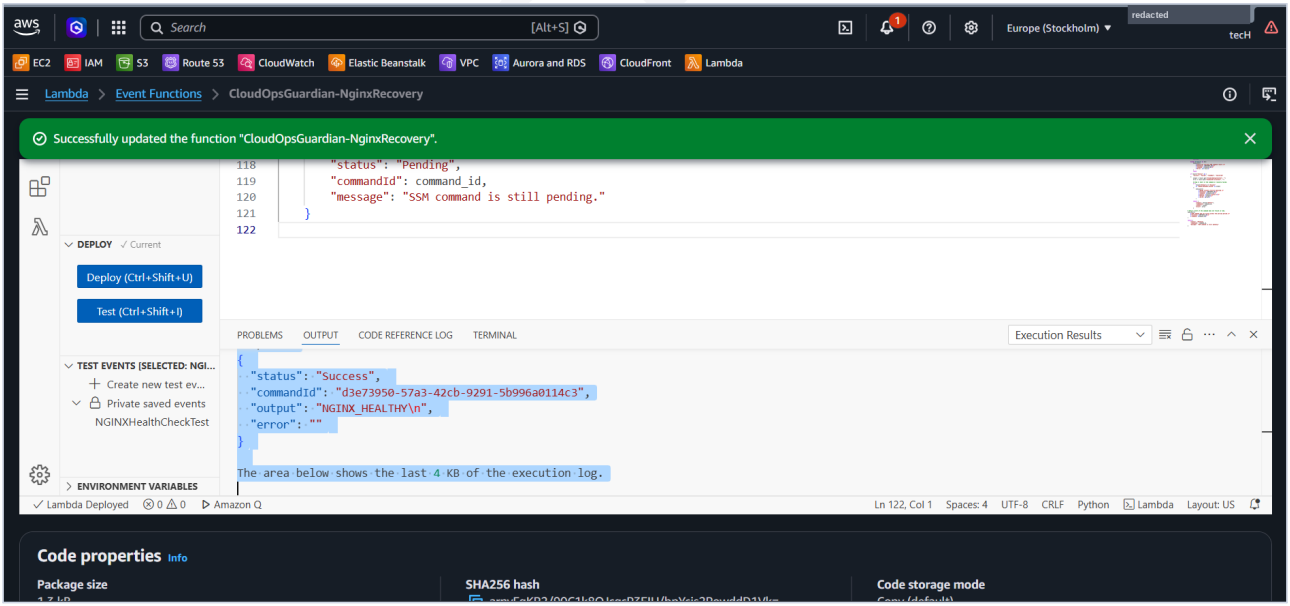
Why Systems Manager is used

Systems Manager Run Command lets Lambda execute a command on a managed EC2 instance without opening an SSH session from Lambda. The EC2 instance must be managed by SSM, the SSM agent must be running, and IAM permissions must allow the relevant actions.

IAM permissions

The Lambda execution role needs permissions for `ssm:SendCommand` and `ssm:GetCommandInvocation`. For email alerts, it also needs `sns:Publish` to the existing topic. The EC2 instance role supports Systems Manager and CloudWatch Agent activity.

> **Operational caution**
>
> The current code polls for up to about 24 seconds plus API execution time. Set the Lambda timeout to at least 35 seconds. Also note that SNS publish errors can cause the Lambda invocation to fail; monitor Lambda errors and configure appropriate alerting for failures that happen before the function can publish to SNS.

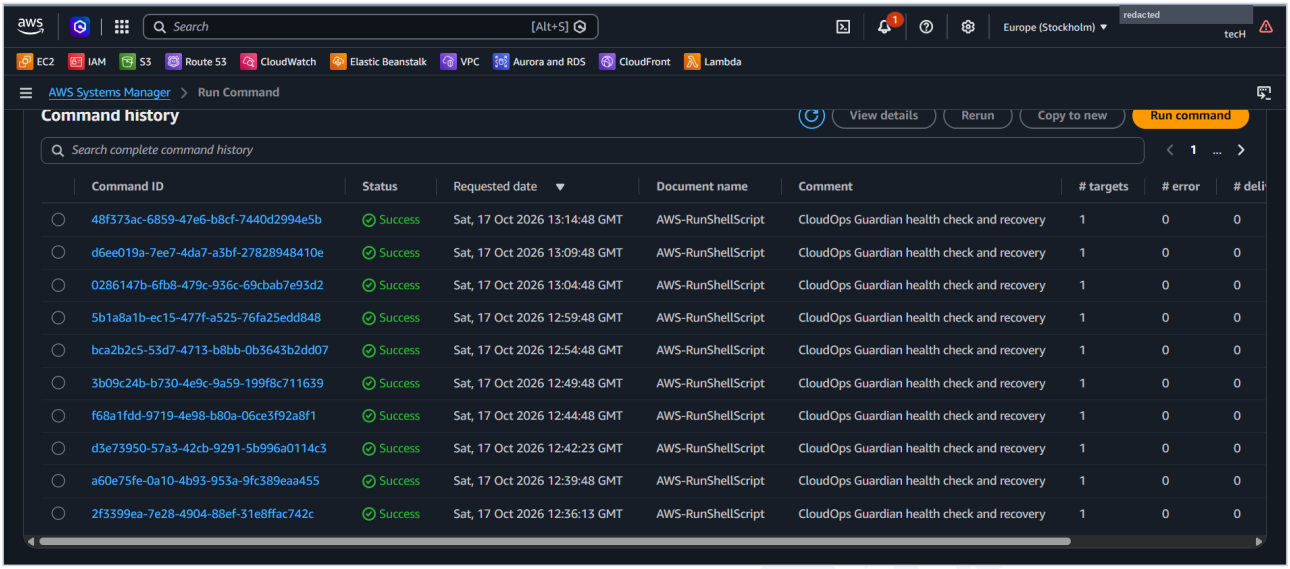


Lambda function CloudOpsGuardian-NginxRecovery: test event returning status Success and NGINX_HEALTHY (account details redacted).

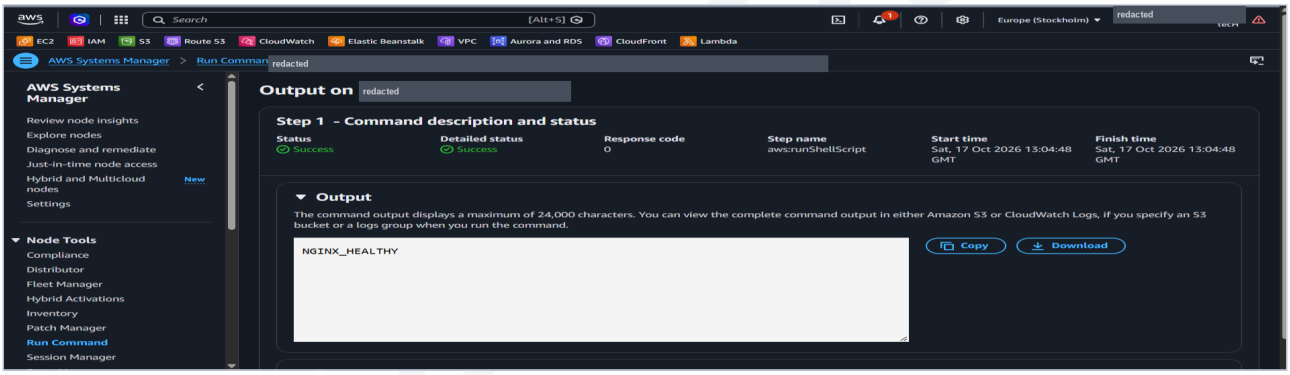
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5. Evidence and Screenshots

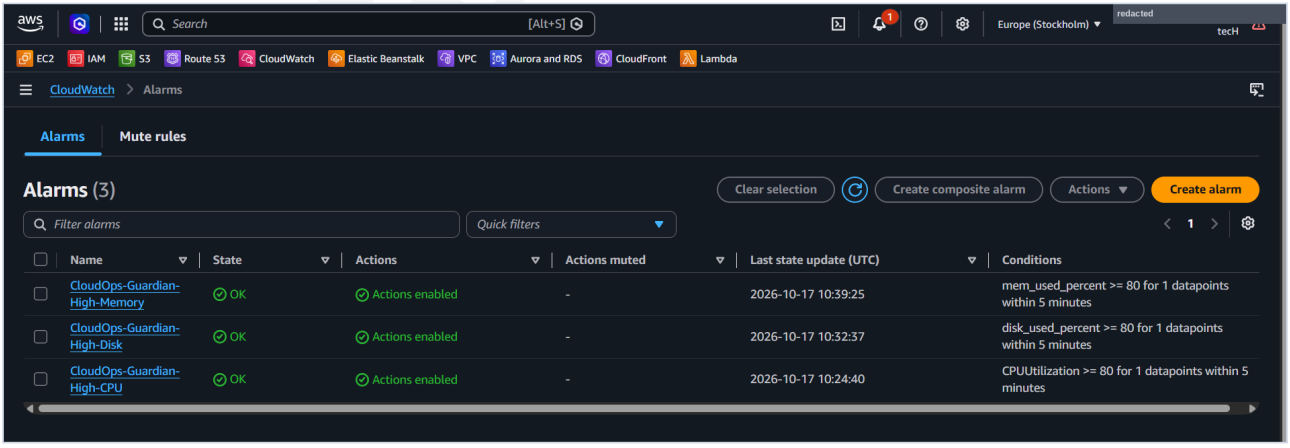
Account IDs, instance and network resource identifiers, the public IP address, and email-related details have been redacted from the screenshots below.



Systems Manager Run Command history: health-check commands with status Success.



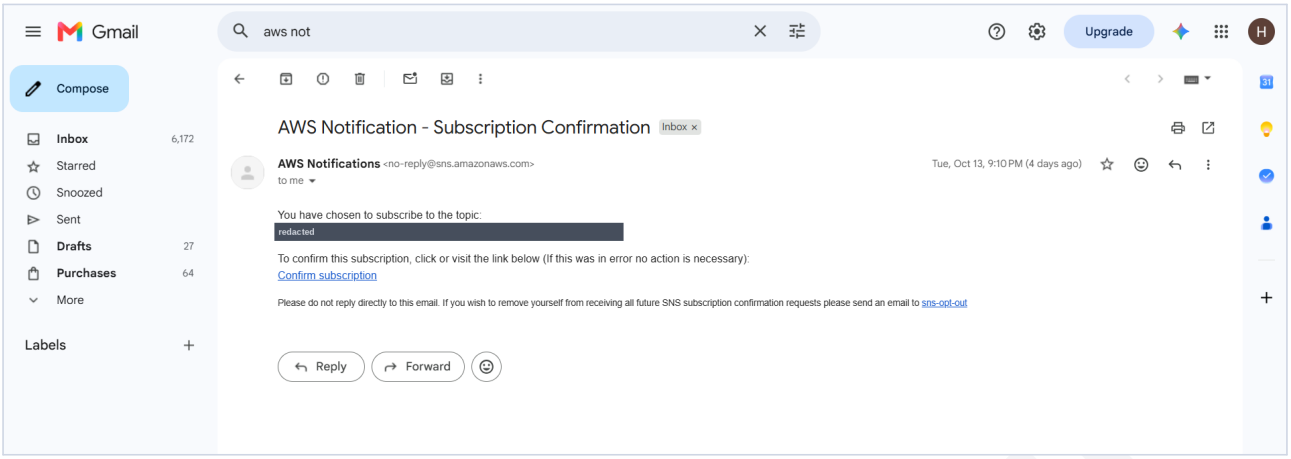
Run Command output for one health check: NGINX_HEALTHY (identifiers redacted).



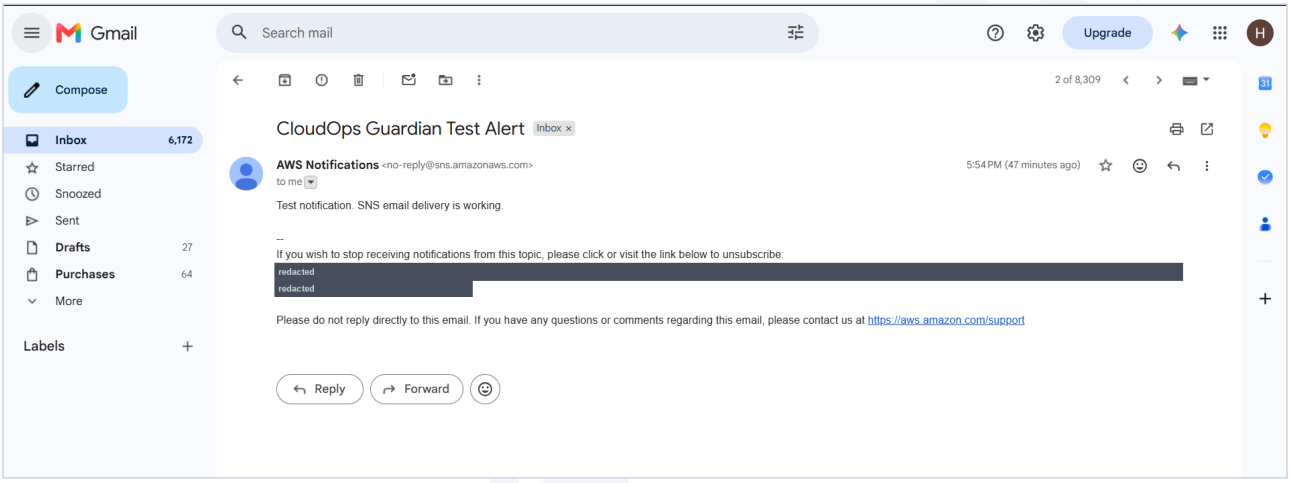
CloudWatch alarms: High-Memory, High-Disk, and High-CPU in state OK.

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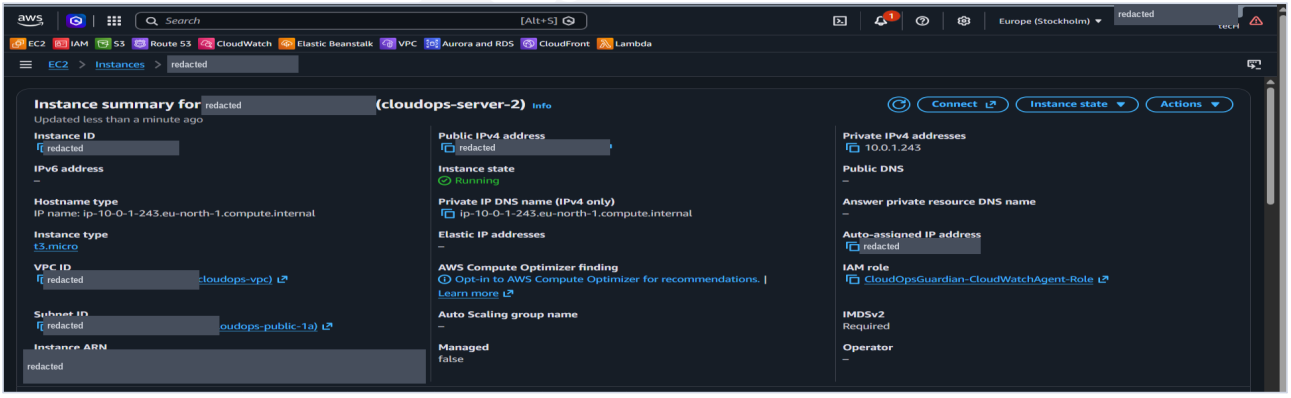
5. Evidence and Screenshots (continued)



SNS subscription confirmation email (topic ARN redacted).



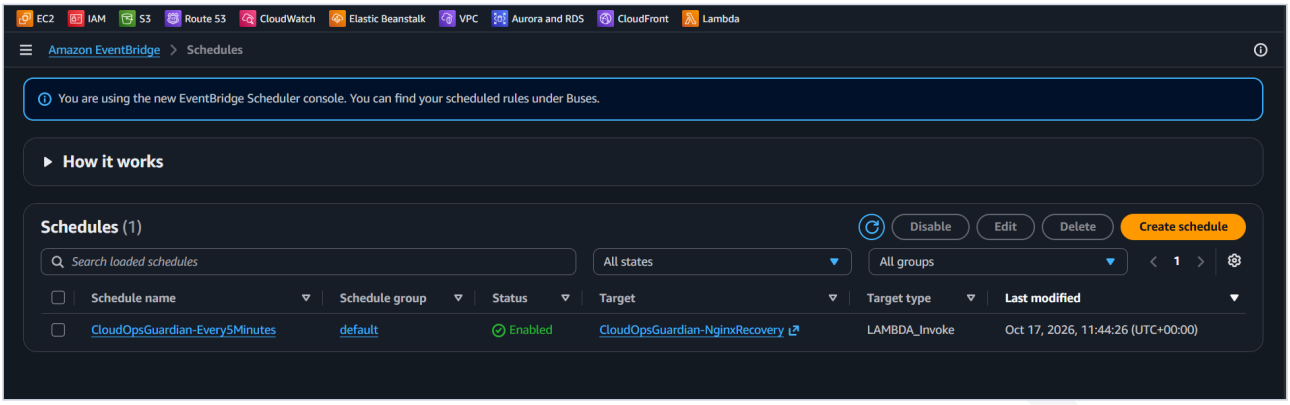
Manual SNS test email delivered (unsubscribe link and endpoint redacted).



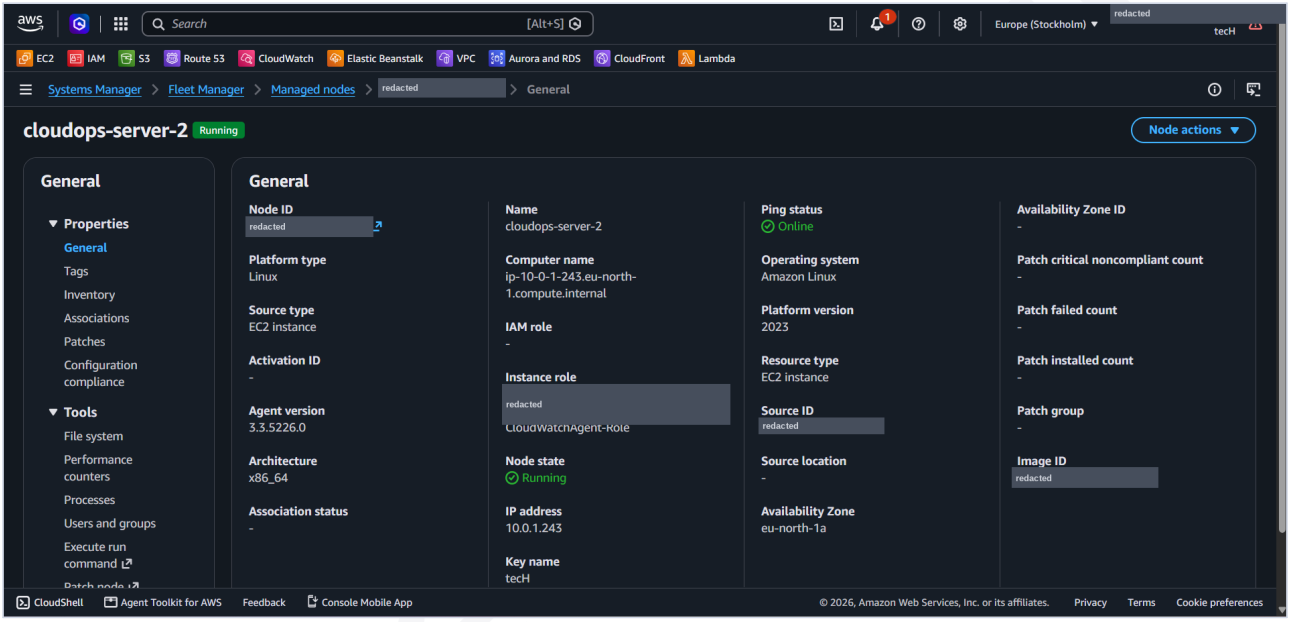
EC2 instance summary: t3.micro in the cloudops-vpc / cloudops-public-1a network (identifiers redacted).

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5. Evidence and Screenshots (continued)



EventBridge Scheduler: CloudOpsGuardian-Every5Minutes, enabled, target is a Lambda invoke.



Systems Manager Fleet Manager: managed node online, Amazon Linux 2023 (identifiers redacted).

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6. Testing Checklist

Check	Evidence / status
Lambda health check	Confirmed: test output returned `NGINX_HEALTHY`.
NGINX auto-recovery	Previously confirmed: output showed `NGINX_DOWN_RESTARTING` and `NGINX_RECOVERY_SUCCESS`.
CloudWatch metrics	Configured: CPU, memory, and disk metrics.
SNS delivery	Confirmed: a manually published SNS test email arrived.
Lambda failure email	Still needs end-to-end verification of an actual failure condition.
Scheduler invocation	Confirm recent EventBridge Scheduler invocations and Lambda logs before claiming scheduled execution is fully verified.

7. Security and Cost Notes

- Restrict SSH inbound access to a trusted IP address, or prefer Systems Manager for administration where practical.
- Apply least-privilege IAM policies; limit SNS publishing to the single alert topic and SSM actions to the required instance/document where supported.
- Do not publish PEM keys, AWS credentials, account-sensitive screenshots, or public IP details in a public repository.
- Check AWS billing regularly. EC2, public IPv4 addresses, CloudWatch metrics/alarms, log ingestion, and other services can incur charges depending on usage and current pricing.
- Keep the recovery function's timeout aligned with its polling period.

8. Authorship, Copyright and Attribution

Creator	Hitesh Patil — created, developed, and documented the CloudOps Guardian project.
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Original project implementation and documentation by Hitesh Patil.

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